Finite volumes on polyhedral cells: exact moments, polynomial reconstruction and the shallow-water equations

Oswald Knoth

Abstract

We describe the finite-volume method of the Fortran code POLYEDER as it is carried over to the Julia package `dmplex`: cell complexes of general polyhedra with vertex coordinates, the exact integration of monomials over cells and faces, the geometry derived from these moments, a polynomial reconstruction that preserves the cell averages, face gradients from interaction regions around the vertices, and the rotating shallow-water equations on a single layer of prisms over the sphere. The moments are computed by homogeneous numerical integration: Euler's identity and the divergence theorem in the affine hull of each cell, face and edge reduce the integral of a monomial to the vertices, following the cone hierarchy of the cell complex, without the degenerate cases of vertex formulas and for non-convex cells. Working draft: Sections 2–5 describe what is implemented; the later sections are the plan.

Keywords: finite volumes, polyhedral meshes, integration of polynomials, k-exact reconstruction, multipoint flux approximation, shallow-water equations

1. Introduction

POLYEDER is a finite-volume code for general polyhedral cells: cut cells over orography, Voronoi and cubed-sphere meshes of a spherical shell, refined and coarsened meshes. Scalars are cell averages, the momentum is the normal component at the faces. Three ingredients carry the method:

1. the polynomial reconstruction of cell averages, for the upwind fluxes of the transport of scalars and momentum, which needs the moments $\int_C (\mathbf{x} - \mathbf{x}_0)^\alpha$ of the cells of a stencil;

2. the gradient and the interpolation to the faces from local problems in the interaction region around each vertex (the corners of the cells at the vertex), a multipoint flux approximation;
3. the velocity at the cells from the normal components at the faces and the projection of the cell tendencies back to the faces.

The shallow-water equations are solved on one layer of prisms over the sphere with free-slip top and bottom, the same code as for the three-dimensional Euler equations.

In `dmplex` the method works on a cell complex (Section 2) that is built from an extruded surface mesh and later from general three-dimensional meshes. The moments are computed by homogeneous numerical integration [6, 3] (Section 3), the geometry of cells and faces from the moments of degree one (Section 4).

2. Cell complexes

A three-dimensional cell complex consists of cells, faces, edges and vertices. The *cone* of a cell is the list of its faces, that of a face the list of its edges, that of an edge its two vertices, each with an incidence number ± 1 of the cellular boundary,

$$\partial C = \sum_{F \in \mathrm{cone}(C)} s_{CF}\, F, \qquad \partial F = \sum_{E \in \mathrm{cone}(F)} s_{FE}\, E, \qquad \partial E = v_2 - v_1, \tag{1}$$

such that $\partial\partial = 0$. The orientation of a face is the circulation of its oriented boundary, its normal follows by the right-hand rule, and $s_{CF} = +1$ when the normal points out of C. A face between two cells has opposite signs in their cones, a face on the boundary of the domain one sign.

`dmplex` stores a surface mesh as a `Plex` (cells, edges and vertices with cones and supports, the analogue of PETSc's `DMPlex`). Its extrusion `ExtrudedPlex` into n layers is the complex of the tensor prisms: a base cell times a layer is a cell, times a level a horizontal face; a base edge times a layer a lateral face, times a level a horizontal edge; a base vertex times a layer a vertical edge, times a level a vertex. Only the surface mesh and n are stored; cones and signs follow from those of the base, $\partial(p \times I) = \partial p \times I + (-1)^{\dim p}\, p \times \partial I$. The `PolyComplex` holds the cones of the three dimensions in compressed form with the vertex coordinates; `extruded_complex` places the vertex v of the base at the level of height z_k at $(a + z_k)\, v$ on a sphere of radius a and at $v + z_k \hat{\mathbf{z}}$ on a plane. A general three-dimensional mesh fills the same arrays.

3. Exact moments of polytopes

3.1. Homogeneous integration

Let $P \subset \mathbb{R}^3$ be a k-dimensional polytope ($k = 1, 2, 3$: a segment, a planar polygon, a polyhedron), $H = \operatorname{aff} P$ its affine hull and f a homogeneous polynomial of degree q, so that $\mathbf{y} \cdot \nabla f = qf$ (Euler). For a point $\mathbf{y}_0 \in H$ the vector field $X = \mathbf{y} - \mathbf{y}_0$ is tangent to H with $\operatorname{div}_H X = k$, and the divergence theorem in H gives

$$\int_P \operatorname{div}_H(Xf) = \int_P \Big(kf + (\mathbf{y}-\mathbf{y}_0)\cdot\nabla f\Big) = (k+q)\int_P f - \int_P \mathbf{y}_0 \cdot \nabla f = \sum_G d_G \int_G f, \tag{2}$$

where G runs over the facets of P with the outward unit normals ν_G in H, and $d_G = (\mathbf{y} - \mathbf{y}_0) \cdot \nu_G$ is constant on G (the distance of $\operatorname{aff} G$ from $\mathbf{y}_0$). For the monomials $\mathbf{y}^\alpha$, $|\alpha| = q$, this is the recursion

$$\boxed{(k + |\alpha|) \int_P \mathbf{y}^\alpha = \sum_G d_G \int_G \mathbf{y}^\alpha + \sum_{i=1}^{3} y_{0,i}\, \alpha_i \int_P \mathbf{y}^{\alpha - e_i}} \tag{3}$$

in the degree for P and down to the vertices in the dimension. With $\mathbf{y}_0$ the foot of the origin on H the last term vanishes for $k = 3$ ($\mathbf{y}_0 = 0$), and on a segment $[\mathbf{a}, \mathbf{b}]$ with the unit tangent $\mathbf{t}$ the facets are its end points with $d_\mathbf{b} = \mathbf{t} \cdot \mathbf{b}$, $d_\mathbf{a} = -\mathbf{t} \cdot \mathbf{a}$ and $\int_{\{\mathbf{v}\}} \mathbf{y}^\alpha = \mathbf{v}^\alpha$. The order-zero moments are the measures: $k\,|P| = \sum_G d_G |G|$, the formula of the volume of a pyramid. For a polygon with the vertices $\mathbf{y}_1, \ldots, \mathbf{y}_m$ in the order of its orientation, the normal $\mathbf{n} = \mathbf{A}/|\mathbf{A}|$ follows from the area vector $\mathbf{A} = \frac{1}{2}\sum_i \mathbf{y}_i \times \mathbf{y}_{i+1}$ (Newell), and the outward normal of the edge from $\mathbf{y}_i$ to $\mathbf{y}_{i+1}$ in its plane is $\nu = \mathbf{t} \times \mathbf{n}$.

Remark 1. POLYEDER integrates the ridge monomials $(\mathbf{b} \cdot (\mathbf{x} - \mathbf{x}_0))^n$ over a cell by a vertex formula with weights of the vertices that depend on $\mathbf{b}$; faces and edges normal to $\mathbf{b}$ need separate cases (`SwitchInt`). The recursion (3) has no such cases, holds for non-convex cells and yields all monomials $\mathbf{y}^\alpha$ of a degree at once; a ridge monomial is a linear combination of them.

3.2. Frames and non-planar faces

The moments of a cell are taken in the local coordinates $\mathbf{y} = (\mathbf{x} - \mathbf{x}_C)/h_C$ with $\mathbf{x}_C$ the mean of its vertices and h_C the largest distance of a vertex from it, so that $|\mathbf{y}| \le 1$ on the cell and the recursion is well conditioned for any position and size of the cell: the returned moments are $\int_C \Big((\mathbf{x} - \mathbf{x}_C)/h_C\Big)^\alpha dV$.

Faces of the cell are integrated in the frame of the cell. A face whose vertices do not lie in one plane — a spherical polygon with more than three vertices — is replaced by the fan of triangles from the mean of its vertices, oriented as the face; both cells of the face see the same triangles, so the complex stays closed. The lateral faces of the prisms are planar, since both of their vertical edges lie on radii.

3.3. Tests

The moments of all monomials up to degree four are exact to round-off on the unit cube, on a triangular prism and on a non-convex L-shaped prism (`test/polyhedral.jl`), as is the divergence theorem $\int_C \partial_i \mathbf{y}^\alpha = \sum_F s_{CF}\, n_{F,i} \int_F \mathbf{y}^\alpha$ with the face moments.

4. Geometry

The volume and the centroid of a cell are its moments of degree zero and one, the area vector of a face is the sum of those of its planar pieces, its centroid follows from the face moments. The outward area vectors of every cell sum to zero (to 10^{-17}, Table 1), which makes the divergence of a constant field vanish exactly. On a spherical shell of thickness d the prisms have flat faces inscribed in the shell; their total volume converges at second order to that of the shell, on triangles, hexagons and the squares of the cubed sphere.

Table 1: Single layer of prisms on the unit sphere, thickness 0.1: relative error of the total volume against $\frac{4\pi}{3}(1.1^3 - 1)$ and the largest $|\sum_F s_{CF}\mathbf{A}_F|$ of a cell.

mesh	cells	volume error	closure
icosahedral, level 2	320	-3.38×10^{-2}	2×10^{-17}
icosahedral, level 3	1280	-8.61×10^{-3}	1×10^{-17}
icosahedral, level 4	5120	-2.16×10^{-3}	5×10^{-18}
hexagonal, level 2	162	-2.66×10^{-2}	3×10^{-17}
hexagonal, level 3	642	-6.76×10^{-3}	1×10^{-17}
hexagonal, level 4	2562	-1.70×10^{-3}	5×10^{-18}
cubed sphere, 4×4	96	-6.51×10^{-2}	6×10^{-17}
cubed sphere, 8×8	384	-1.67×10^{-2}	2×10^{-17}
cubed sphere, 16×16	1536	-4.21×10^{-3}	9×10^{-18}

5. Polynomial reconstruction

5.1. Average-preserving least squares

For every cell C we seek the polynomial $p_C = \sum_k c_k \, \mathbf{y}^{\alpha_k}$ of degree $\le r$ in the local coordinates $\mathbf{y} = Q_C(\mathbf{x} - \mathbf{x}_C)/h_C$ ($\mathbf{x}_C$ the centroid, h_C the largest distance of a vertex, Q_C orthogonal) that has the average $\bar{u}_C$ of the cell and fits the averages of the other cells of its stencil S_C in the least-squares sense [2, 7],

$$\frac{1}{|C|}\int_C p_C = \bar{u}_C, \qquad \sum_{C' \in S_C \setminus C} \Big(\frac{1}{|C'|}\int_{C'} p_C - \bar{u}_{C'} \Big)^2 \to \min . \tag{4}$$

With $a_{C'k} = |C'|^{-1} \int_{C'} \mathbf{y}^{\alpha_k}$ (the moments of Section 3 in the frame of C) the constraint eliminates the constant, $c_0 = \bar{u}_C - \sum_{k\ge1} a_{Ck} c_k$, and the remaining coefficients solve

$$\sum_{k\ge1} (a_{C'k} - a_{Ck})\, c_k = \bar{u}_{C'} - \bar{u}_C, \qquad C' \in S_C \setminus C, \tag{5}$$

by the pseudo-inverse with a relative rank tolerance (POLYEDER's `TolRank`). The coefficients are linear in the averages of the stencil, $c = R_C \, \bar{u}_{S_C}$, computed once.

Stencils. S_C is C with the cells that share a vertex with it, grown ring by ring until the reduced matrix has full rank and at least 1.5 times as many rows as monomials. Inside a mesh two rings suffice for degree three; at boundaries the stencil grows, for instance in the bottom and top layers of a column, where two rings see only three layers.

Tangential basis. On one layer of cells over the sphere the stencil does not resolve the radial structure, and the radial monomials make the least-squares problem singular. The option `basis = :tangential` takes Q_C with the radius at the centroid as its third axis and uses only the monomials without the third coordinate: a reconstruction in the tangent plane, the moments still exact over the prisms.

Face values. The average of p_C over a face F of C is $|F|^{-1} \int_F p_C = \sum_k c_k \, |F|^{-1} \int_F \mathbf{y}^{\alpha_k}$; for the two cells of every face these are two sparse matrices from the cell averages to the face values (`face_value_matrices`), and at the points of a face quadrature two more (`point_value_matrices`). On a distorted triangle mesh of a box, extruded into five layers, all face values of polynomials of degree ≤ 3 are reproduced to 10^{-12}.

5.2. Transport

The transport $\partial_t q + \nabla \cdot (q\mathbf{u}) = 0$ of the cell averages takes the upwind values at the faces. With one volume flux $U_F = \int_F \mathbf{u} \cdot \mathbf{n}$ per face, as for the normal velocities of a staggered scheme, the flux is $U_F\, \bar{q}_F$ with the upwind face average. This is only second order, whatever the degree: $\int_F q\, \mathbf{u} \cdot \mathbf{n} - U_F \bar{q}_F = O(h^2)\, |F|$ unless $\mathbf{u} \cdot \mathbf{n}$ is constant on F. With the velocity field known at the points $\mathbf{x}_g$ of a face quadrature (`face_quadrature`: the seven-point rule of degree five [4] on the fan of triangles of the face) the flux $\sum_g w_g (\mathbf{u} \cdot \mathbf{n})_g\, q_g$, upwinded point by point, has the order $r + 1$ of the reconstruction. Time integration is the third-order SSP Runge–Kutta method.

Test fluxes that are exactly divergence free in every cell follow from a vector potential: for the velocity $\mathbf{u} = g(\eta) \nabla\psi \times \nabla\eta$ of a stream function ψ of the horizontal position (η the radius on the sphere, the height z on a plane) the flux through F is the circulation of $g(\eta)\psi\nabla\eta$ around F, $U_F = \sum_{a \to b} \frac{1}{2}(\psi(a) + \psi(b))\, (G(\eta_b) - G(\eta_a))$ over the oriented edges of F with $G' = g$; it vanishes on the faces of constant η, and the fluxes of a cell sum to zero exactly. On the sphere $\psi = \boldsymbol{\Omega} \cdot \hat{\mathbf{r}}$ with $G(\eta) = \eta^3/3$ gives the solid-body rotation $\boldsymbol{\Omega} \times \mathbf{x}$.

Table 2: Steady flow $\mathbf{u} = (\sin \pi x \cos \pi y, -\cos \pi x \sin \pi y)$ in the unit square, distorted triangles ($2n^2$ cells, one layer), $q = \exp(2\psi)$, $\psi = \sin \pi x \sin \pi y / \pi$, relative L^2 error at $t = 0.5$, fluxes at the quadrature points; CFL 0.4.

	degree 1		degree 2		degree 3	
n	error	order	error	order	error	order
8	9.57×10^{-3}		4.98×10^{-3}		2.71×10^{-3}	
16	2.09×10^{-3}	2.20	6.97×10^{-4}	2.84	2.16×10^{-4}	3.65
32	4.21×10^{-4}	2.31	9.69×10^{-5}	2.85	1.05×10^{-5}	4.37
64	9.76×10^{-5}	2.11	1.27×10^{-5}	2.93	7.30×10^{-7}	3.84

With the face fluxes $U_F \bar{q}_F$ the same flow converges at second order for all degrees (1.2×10^{-4}, 1.0×10^{-4}, 1.3×10^{-4} at $n = 64$). On one layer over the sphere (thickness 0.01) a Gaussian hill $\exp(-|\hat{\mathbf{r}} - \hat{\mathbf{r}}_0|^2/0.4^2)$ is rotated once about an axis tilted by 45° with the divergence-free face fluxes (Table 3); mass is conserved to 10^{-14}. Besides the flux, the flat faces of the prisms inscribed in the shell limit the order to two.

Remark 2. For the shallow-water equations with the normal momentum at the faces, the mass flux through a face is a single number, and the transport

Table 3: Solid-body rotation of a Gaussian hill over one revolution on one layer of prisms over the unit sphere, tangential basis, face fluxes $U_F \bar{q}_F$: relative L^2 error.

mesh	cells	degree 0	degree 1	degree 2	degree 3
hexagonal 3	642	7.6×10^{-1}	3.3×10^{-1}	3.7×10^{-1}	1.3×10^{-1}
hexagonal 4	2562	6.3×10^{-1}	1.2×10^{-1}	1.2×10^{-1}	2.2×10^{-2}
hexagonal 5	10242	4.7×10^{-1}	3.3×10^{-2}	2.2×10^{-2}	4.5×10^{-3}
icosahedral 3	1280	6.1×10^{-1}	2.3×10^{-1}	1.4×10^{-1}	9.8×10^{-2}
icosahedral 4	5120	4.6×10^{-1}	7.0×10^{-2}	3.1×10^{-2}	2.0×10^{-2}
icosahedral 5	20480	3.1×10^{-1}	1.8×10^{-2}	5.7×10^{-3}	4.6×10^{-3}

is second order like that of every staggered scheme; the higher reconstruction lowers the error constant and the dispersion (Table 3). Higher order needs the normal velocity along the face (from the reconstructed cell velocity, or several unknowns per face) and cells with curved faces on the sphere.

6. Interaction regions and the face gradient

Plan. Around each vertex the corners of its cells (a cell at the vertex with the faces meeting there) form the interaction region of a multipoint flux approximation [1]: a linear gradient per corner, continuity of the values at the faces of the region and the cell averages give a small local system; its solution expresses the gradient and the value at every face in the averages of the cells around its vertices (POLYEDER: `TGrad`).

7. The shallow-water equations

Plan. One layer of prisms over the sphere, free-slip top and bottom. Height h as cell averages, the normal momentum $(h\mathbf{u}) \cdot \mathbf{n}$ at the lateral faces. The pressure gradient $g\nabla(h^2/2)$ (with the bathymetry) at the faces from Section 6; the cell velocity from the normal components [8], its advection with the reconstruction of Section 5 and the Coriolis force at the cells, the tendency projected back to the faces. Time integration with the third-order TVD Runge–Kutta method, later with split-explicit methods [9]. Tests: Williamson's case 2 [10] and the barotropic instability of Galewsky et al. [5].

Code availability

`src/Polyhedral/` of `dmplex`: `Complex.jl` (`PolyComplex`, `extruded_complex`), `Integration.jl` (`MonomialBasis`, `segment_moments`, `polygon_moments`, `face_moments`, `cell_moments`), `Geometry.jl` (`PolyGeometry`), `Reconstruction.jl` (`PolyReconstruction`, `face_value_matrices`, `point_value_matrices`), `Advection.jl` (`PolyAdvection`, `stream_function_flux`); tests in `test/polyhedral.jl`.

References

[1] I. Aavatsmark, An introduction to multipoint flux approximations for quadrilateral grids, *Comput. Geosci.* 6 (2002) 405–432.

[2] T. J. Barth and P. O. Frederickson, Higher order solution of the Euler equations on unstructured grids using quadratic reconstruction, AIAA paper 90-0013 (1990).

[3] E. B. Chin, J. B. Lasserre and N. Sukumar, Numerical integration of homogeneous functions on convex and nonconvex polygons and polyhedra, *Comput. Mech.* 56 (2015) 967–981.

[4] D. A. Dunavant, High degree efficient symmetrical Gaussian quadrature rules for the triangle, *Int. J. Numer. Methods Eng.* 21 (1985) 1129–1148.

[5] J. Galewsky, R. K. Scott and L. M. Polvani, An initial-value problem for testing numerical models of the global shallow-water equations, *Tellus* 56A (2004) 429–440.

[6] J. B. Lasserre, Integration on a convex polytope, *Proc. Amer. Math. Soc.* 126 (1998) 2433–2441.

[7] C. Ollivier-Gooch and M. Van Altena, A high-order-accurate unstructured mesh finite-volume scheme for the advection–diffusion equation, *J. Comput. Phys.* 181 (2002) 729–752.

[8] B. Perot, Conservation properties of unstructured staggered mesh schemes, *J. Comput. Phys.* 159 (2000) 58–89.

[9] J. Wensch, O. Knoth and A. Galant, Multirate infinitesimal step methods for atmospheric flow simulation, *BIT Numer. Math.* 49 (2009) 449–473.

[10] D. L. Williamson, J. B. Drake, J. J. Hack, R. Jakob and P. N. Swarztrauber, A standard test set for numerical approximations to the shallow water equations in spherical geometry, *J. Comput. Phys.* 102 (1992) 211–224.